Sustainable Chemistry and Pharmacy 32 (2023) 100999



Contents lists available at ScienceDirect

Sustainable Chemistry and Pharmacy

journal homepage: www.elsevier.com/locate/scp





The use of biochar made from biomass and biosolids as a substrate for green infrastructure: A review

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ARTICLE INFO

Handling Editor: Klaus Kümmerer

Keywords:
Green infrastructure
Green roof
Growing medium
Biomass
Biosolids
Pyrolysis
Biochar

ABSTRACT

Biochar made from biomass and biosolids can be used in substrates suitable for urban green infrastructure, especially green roofs, green parking lots, and green walls. The properties of biochar depend on the feedstock and pyrolysis conditions. Biochar made from different feedstocks, e.g., sludges, biomass, food waste, wood, and plant residues, has different physicochemical properties that affect the availability of nutrients, absorption of pollutants and harmful gases, carbon sequestration, and improvement of pH of amended soil. This paper reviews and summarizes various feedstocks for biochar production, pyrolysis technology and conditions, and physicochemical properties of selected types of biochar. Since biochar has a positive influence on plant growth, fertilizer efficiency, and rainwater capture, showing its use as a stable organic component of the substrate is the main objective of this research paper. Conclusion of the review considers the usage of biochar as a substrate for green infrastructure in pursuance of its characteristics, limiting pollutants concentration, potential benefits and risks for the environment, water sources and occurring climate change. The review is based on the analysis of selected papers on existing research and case studies related to the use of biochar and the evaluation of the impact of biochar on the characteristics of growing medium.

1. Introduction

This paper evaluates knowledge of the current state of wastewater management systems and biodegradable waste management towards the circular economy paradigm and sustainability as targeted in The European Green Deal, one of the Commission's priorities for 2019–24 (The European Green Deal, 2019). Fundamental to waste management is the EU circular economy action plan or the bioeconomy strategy and the "Farm-to-Fork" strategy, where the main objective is the recovery of nutrients (P, N). The use of primary resources including water is increasing with the growing population which has an impact on the accruing production of wastewater (WW) and its by-product after treatment - biosolids. The increasing volume of biosolids is exceeding the limit capacities and the possibility of its further incineration, landfilling or reuse. Regulations for the member countries of the Europe Union (EU) regarding the use of biosolids in agriculture are based on the Council Directive 86/278/EEC from 1986 (Council Directive 86/278/EEC, 1986). Monitoring, usage, and disposal of biosolids are based on the Council Directive 91/271/EEC from 1991 (Council Directive 91/271/EEC, 1991). Both directives have as their main aim the protection of the environment and human health while using or disposing of

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https://doi.org/10.1016/j.scp.2023.100999
Received 29 November 2022; Received in revised form 11 January 2023; Accepted 15 January 2023
Available online 1 February 2023
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biosolids. The increased production of biosolids and the limitation of their disposal led to finding alternative solutions for recycling and further usage.

The implementation of management change towards the circular WW management is specified in the framework EC communication COM/2015/0614 final (COM, 2015). Further details are provided in the national current practices and legislation constraints that are specific for each individual country. In terms of biosolid reuse and nutrition potential from the WW treatment several limits exist, namely, the restriction for the use of biosolids on agricultural soil. It is now known that in addition to heavy metals, organic micropollutants (such as drugs and their residue), personal care products, polyaromatic hydrocarbons (PAHs), perfluoroalkyl substances (PFAS), and microplastics) have to be considered. The commercial opportunity has been introduced by Directive 2009/28/ES (Directive 2009/28/ES, 2018) on the promotion of the use of energy from renewable sources. However, modern regulatory principles based on risk assessment, respect the cost sustainability of the application of sewage sludge and nutrients, evaluate the agrochemical availability of phosphorus, etc.

The new EU Fertilizer Products Regulation (EU) 2019/1009 (Regulation 2019/1009, 2019) expands the harmonization legislation for making EU fertilizer products available on the European Single Market and in particular extends the scope to include soil improvers, growing media, and plant biostimulants. There is also a list of materials that can be used as ingredients in the manufacture of fertilizer products (including limits on contaminants), labeling rules, and how compliance with the requirements of the regulation will be demonstrated.

A similar process applies to other organic wastes mainly produced from the food and agriculture sectors, known as biomass. Reusable biomass includes waste from agricultural production, wood waste, animal manure, and biomass from biodegradable municipal waste. Almost all wastes of biological origin can currently be reprocessed and reused. Such an ecological circular system leads to a positive impact on the environment as well as ongoing climate change (Rizos et al., 2017). In this paper, we focus on biochar produced from biosolids, food industry, paper and pulp production, straw waste, nuts shells, digested fish waste, and animal manure.

Biochar is obtained as a possible product by pyrolysis (combustion at low oxygen conditions) of various feedstock types (Ronsse et al., 2013; Bolognesi et al., 2021; Callegari et al., 2018). The resulting solids are rich in carbon and other elements and when admixed with soil, several positive aspects occur such as improved nutrient immobilization (Wang et al., 2021; Hossain et al., 2020), water retention (Basso et al., 2013; Wang et al., 2014), and control of contaminants (He et al., 2019; Ji et al., 2022). For this reason, the addition of biochar to soil is proposed as an appropriate strategy not only for soil quality improvement but also as a means for the remediation of contaminated soils. Carbon storage in the form of biochar also contributes to the reduction of carbon dioxide (greenhouse gas) release. The use of biochar can therefore present an effective strategy to cope with the decreasing quality of agricultural land, the high demands on food self-sufficiency, the risks of soil contamination, and the threat of the greenhouse effect (Mandal et al., 2016a).

Green infrastructure (GI) is one of the key adaptation measures to climate change (Green Infrastructure and Climate Adaptation, 2014). It is a combined network of urban greenery and local rainwater management. In an urban environment, GI combines technical solutions with natural principles into a functional whole (Wang and Banzhaf, 2018). An important prerequisite for healthy growth and resilience of urban greenery is the use of suitable substrates that are capable of high water retention due to porosity and at the same time provide a friendly environment for plant root growth ("Landscape fragmentation in Europe", 2011). According to its unique properties, biochar becomes a key functional element of all substrates. Applications of biochar in the soil of GI systems improve the soil properties. Biochar improves physical (e.g., water holding capacity, greenhouse gases (GHGs) removal and moisture level), chemical (e.g., pollutants immobilization and carbon sequestration), and biological (e.g., nutrients, microbial abundance, and diversity) properties of the soils (Dai et al., 2013; Joseph et al., 2018; Gul et al., 2015). The GI concept leads to a reduction in the temperature in buildings during summer (green roofs (GRs) and walls) and reduces the energy needs for cooling buildings, thus improving the microclimate in cities (Santamouris, 2014). The GI concept will capture rainfall and reduce surface runoff into the sewer (Li et al., 2019), nutrients in biochar (N, P, C, K) will be used in the GI and the idea of "waste to product" will be implemented (Yang et al., 2020). In the case of green parking lots, traffic pollution (petrol, oil) will be reduced by adsorption on the biochar (Schifman et al., 2018).

When evaluating the effect of biochar on GI structural substrates, it is important to consider the entire process from its production to its deposition in the soil. This is because the production process can affect the properties of the biochar. The scheme in Fig. 1 shows

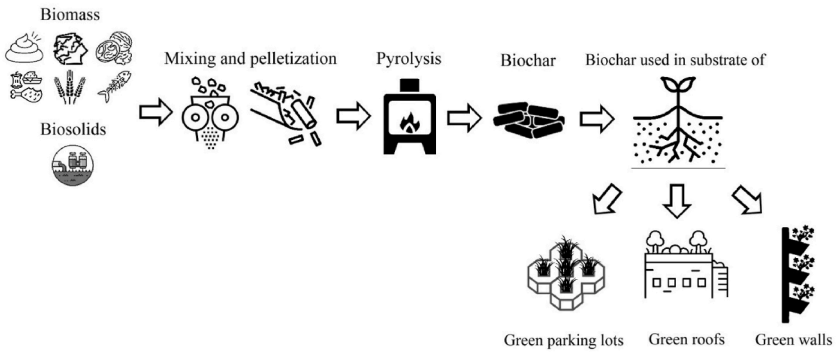


Fig. 1. Scheme of biochar production and use in green infrastructure.

biochar production and its use in GI. First, the input material is finely crushed and then pelletized, then the pellets are pyrolyzed, and biochar is formed. Biochar is then mixed into the substrate medium, which is used, for example, in GRs, green parking lots, and green walls. The biochar production process should always be optimized to ensure that the resulting biochar properties effectively improve the physicochemical properties of the substrates.

The review is based on the analysis of different existing research and case studies related to the use of biochar to evaluate the impact of biochar on substrate characteristics. At the beginning of this review, different types of organic solid waste (biomass and biosolids) are compared with respect to the total amount produced and their physicochemical composition. The pyrolysis process (thermal pyrolysis and microwave torrefaction) and the procedures that precede it (mixing with additives and pelletizing) are also explained. Selected physicochemical properties of the resulting pyrolysis product (biochar) are described and compared depending on the feedstock used. Biochar certification is defined, which determines the classification of biochar into different use categories according to its physicochemical properties. The last chapter describes selected effects of biochar on the GI substrate (increase soil fertility, improve water holding capacity, support soil remediation, carbon sequestration, and mitigation of GHG).

2. Organic solid waste

The process of producing biochar of sufficient quality is based on the pyrolysis of organic biodegradable feedstocks, which are usually found as various types of wastes or by-products. Each of them may have different physical and chemical properties which affect the final quality and beneficial characteristics of resulting biochar (Das et al., 2018). Several types of feedstocks can be used for biochar production. The feedstocks were selected based on an extensive study by Ippolito et al. (2020), which compiled more than 5 400 peer-reviewed journal articles and found that the most commonly used organic feedstocks for pyrolysis include biomass received from food waste (FW), paper waste (PW), straw waste (SW), walnut shells (WS), digested fish waste (DFW), animal manure, and biosolids received from wastewater treatment plants (WWTPs). These feedstocks are also widely used in EU countries, so their use for biochar production and subsequent use in GI is possible. In addition, our team has experience with pyrolysis and utilization of these feedstocks.

Waste production continues to grow relative to income, as do the economic and environmental costs associated with waste disposal. Hence the need to speed up the process of reducing waste production and disposal (Mazzanti and Zoboli, 2008). Fig. 2 shows the production of selected organic waste from 27 EU member states in the period from 2010 to 2020. It should be noted that the volume of sludge produced from WWTP is still increasing and the price for its disposal is also increasing with it (Domini, 2022).

Information on the elemental composition of the feedstock for the pyrolysis process is essential for the production of high-quality biochar with the desired properties. According to the selection of the feedstock, we choose the feedstock pretreatment method and the pyrolysis process (Kambo and Dutta, 2015). The elemental composition of some types of feedstocks is shown in Table 1, the data presented was obtained as an average of samples from different publications.

2.1. Biomass

According to Directive 2018/2001 (Directive (EU) 2018/2001, 2022) on the promotion of the use of energy from renewable sources, biomass is defined as the biodegradable fraction of products, waste, and residues from agriculture, biological origin including plant and animal substances, from forestry and related industries including fisheries and aquaculture, as well as the biodegradable fraction of industrial and municipal wastes. Biomass waste is currently disposed of mainly through landfill storage, incineration, or composting which is the cheapest and easiest option. According to the European Commission (EC), landfilling is the worst method of the disposal of biodegradable waste. This reason led to the adoption of the legislative proposal aimed at phasing out the disposal of recyclable wastes (including biowaste) in landfills, corresponding to a maximum landfilling rate of 10% by 2035 (Council Directive 1999/31/, 1999), and the reduction of biodegradable waste in municipal landfills up to 30% by 2025 (Biodegradable waste, n.d.).

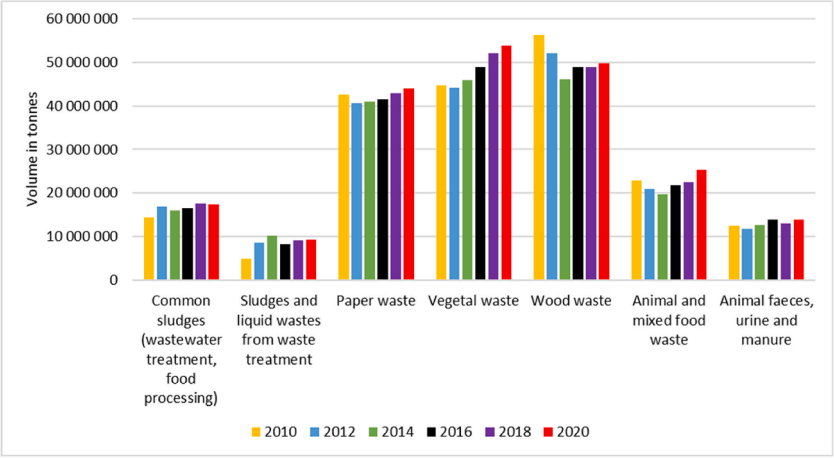


Fig. 2. Generation of selected organic wastes in EU (Generation of waste by waste category, hazardousness and NACE, 2022).

Table 1
Chemical composition of selected raw feedstocks.

Feedstock	pH (–)	C (%)	H (%)	N (%)	O (%)	S (%)	P (%)	Na ($g \cdot kg^{-1}$)	Mg ($g \cdot kg^{-1}$)	Ca ($g \cdot kg^{-1}$)	Authors and notes
Biosolids	6.7	25.9	4.2	3.8	27.3	1.2	8.8	1.8	3.9	28.6	Average of 22 samples (Yang et al., 2018; Zielińska et al., 2015; Gao et al., 2020; Frišták et al., 2018; Fei et al., 2019; Gong et al., 2014)
Food waste	4.9	47.7	6.8	3.0	35.2	0.3	0.4	13.5	1.1	11.3	(Selvam et al., 2021) Average of 102 samples
Paper sludge	7.1	35.6	5.0	4.3	31.8	2.0	0.4	1.2	0.4	0.9	Average of 10 samples (Englatina and Chirwa, 2021; Xu and Lancaster, 2008; Abdullah et al., 2015)
Wheat Straw	6.7	43.5	6.0	0.5	40.1	0.2	–	0.3	1.5	1.5	Average of 5 samples (Singhal et al., 2021; Defoort et al., 2019; Li et al., 2007; Bradna et al., 2016; Cuiping et al., 2004)
Walnut shell	–	48.3	6.2	0.5	45.2	0.1	–	0.6	1.3	9.1	Average of 5 samples (Gupta et al., 2019; ONAY et al., 2004; Kar, 2011; Diao et al., 2021; Albatrni et al., 2022)
Digested fish waste	6.7	56.2	8.7	5.0	22.8	0.9	1.6	7.6	2.1	95.4	Average of 10 samples (Prüter et al., 2020; Ahuja et al., 2020; Vivekanand et al., 2018)
Chicken manure	7.4	38.6	4.6	4.1	33.2	0.5	–	4.1	4.0	11.8	Average of 8 samples (Keskin et al., 2019; Tańczuk et al., 2019; Das et al., 2009; Jindo et al., 2014; Oyewusi et al., 2021; O'Connor, 2004)

2.1.1. Food waste

Food is an essential need for mankind; however, its residues have become a worldwide challenge, that the planet's environment is facing (Malhotra and Vos, 2021). According to the research project UNEP Food Waste Index 2021 (Stenmarck et al., 2016), in 2019, a total of 931 million tons of FW were generated, including food and inedible parts associated with food. This amount includes 61% of FW from households, 26% from services, 13% from retail. FW is mainly composed of the following chemical substances: carbon, oxygen, nitrogen, lipids, proteins, carbohydrates, hemicellulose, cellulose, lignin, non-structural carbohydrates, starch, free sugar, sucrose, and glucose. FW consisting of rice and vegetables is rich in carbohydrates, while FW made from meat and eggs contains a large amount of protein and lipids (Paritosh et al., 2017). Egg shells contain high levels of calcium carbonate ($CaCO_3$), which could be used as an alkaline compound to remove heavy metals (HM) from WW (Ghosh et al., 2016).

2.1.2. Paper waste

Paper production is the sixth-largest polluting industry among production sectors (Ali and Sreekrishnan, 2001). The production of PW and pulp mill residues increases year by year with the increased usage of paper-made products in different industrial areas (Mandal et al., 2016a). According to the Key statistic report of CEPI (2020) (Key Statistics, 2021), the total amount of paper, pulp, and board production in 2020 was approximately 121.4 million tons, while the total amount of paper collected and recycled was 54.5 million tons. Europe's (27 member states of EU + Norway + Switzerland + United Kingdom) total PW and pulp recycling rate in 2020 was established at 73.9%. Sludge generated by paper and pulp mill production contains 60–94% of organic matter (OM) (cellulose or inorganic adhesives) and mineral fillers ($CaCO_3$ or aluminosilicate minerals), although, it includes less nitrogen and phosphorus in contrast to biosolids or compost. The chemical composition of PW consists 20% of hemicellulose, 18% of lignin, 33% of cellulose, and 29% other (Souza et al., 2019). Therefore, nutrient addition is usually needed when PW is further used as a soil supplement in the agriculture sector (Mandal et al., 2016a; Park et al., 2011; Ding et al., 2018).

2.1.3. Straw waste

Straw, one of the most frequented biomass residues, is an agricultural by-product obtained after main product harvesting, generated and collected directly at the field, and used as a feedstock in bio-based production or for other purposes (Rice straw and Wheat straw, 2013). SW is dry stalk residue of various cereals, including straws of wheat, rice, maize, barley, oats, rye, etc. The total residue production of listed SWs in the EU represented an abundant quantity of 294 million tons in 2013 (Searle and Malins, 2013). SW is a complex lignocellulose material that contains 28–39% of cellulose, 23–25% of hemicellulose, and 15–25% of lignin, with lower percent of ashes and proteins (Carvalheiro et al., 2009; Mullen et al., 2015).

2.1.4. Walnut shells

Large quantities of shells from walnuts become a by-product residue considered as an agricultural waste every year worldwide. According to statistics from 2021 (Walnut production, 2021), the largest producer of walnuts is considered China. In 2021 the production of in-shell walnuts in China reaches 1.1 million metric tons. The main importers for Europe are considered the United States and Chile, while the main producers of walnuts in Europe are Romania, France, Spain, and Italy (Medina and Kuypers, 2021).

WSs are part of the lignocellulosic waste biomass group. Lignocellulosic biomass consists of dry matter from plants or animal residues and is considered the most available material used for the generation of energy materials, namely biofuel and bioethanol (Uzun and Yaman, 2017). Lignocellulose is composed of carbohydrate polymers such as cellulose, hemicellulose, and lignin. The chemical composition of WSs corresponds to the values 35% of lignin, 30% of hemicellulose, 20% of cellulose, and 15% others (Morales et al., 2021). Accordingly, WSs are considered to be a proper feedstock for further conversion into bio-valuable chemical and fuel products (Exporting walnuts to Europe, 2019).

2.1.5. Digested fish waste

DFW has become an important and cheap source of bio-based production. Several alternative strategies have been developed to reduce a large amount of DFW produced worldwide. Two different methods for improving the economic value of DFW have been developed, namely mass transformation and sorting. The mass transformation includes the conversion of DFW into fish meal, fish oil, fertilizers, etc. (Sharp and Mariojouls, 2012). DFW is suitable for agricultural use owing to its high contents of nutrients, such as N, P, and Ca (Radziemska et al., 2019). According to the results of the research, Shikhaliyev et al. (2018) concluded that the cost-effective production, low metal content, and easily degradable character of biochar catalyst produced from DFW present a potential catalyst for common industrial applications.

2.1.6. Manure waste

A present decrease in the availability of ground for the production of crops is reflecting the increase in animal production demand for mankind's daily consumption. Hence, demand is likely to be met mainly through massive poultry and livestock production (Thangarajan et al., 2013). Manure produced by animals is an essential material used in agriculture due to the content of significant elements and nutritional value necessary for plant growth. The highest quantities of manure come mainly from poultry production and livestock. In the EU is produced about 1400 million tons of livestock manure (Flotats et al., 2013). Greater production of animal manure (AM) is associated with several additives in the feed, including iron (Fe), cobalt (Co), arsenic (As), manganese (Mn), copper (Cu), and selenium (Se). Stated additives help to improve weight gain, reduce diseases, and improve the generation of dairy and eggs. However, feed amendments rich in metals serve as a direct source of contamination (Mandal et al., 2016a; BOLAN et al., 2004).

2.2. Biosolids

Biosolids are a by-product after the whole treatment process. Municipal WW is treated before it is discharged into the environment, and there are three primary stages of treatment. Primary treatment includes the removal of grit and sedimentation, while secondary treatment involves the oxidation of dissolved OM by utilizing biologically active sludge. Tertiary treatment employs more advanced biological methods of nitrogen removal and physical and chemical processes (Sonune and Ghate, 2004). The main composition of biosolids is water, OM, minerals, and nutrients (Raček et al., 2022; Rorat et al., 2019). Treatment processes that remove solids in WWTPs include screenings and grit, naturally floating scum, and the solids removed during primary and secondary treatment. Biosolid is suitable for recycling and further reuse, mostly as a soil amendment. Biosolids, on the other hand, have become a term referring to untreated primary and secondary organic solids. The difference in terms of biosolids refers to the organic solids that have received stabilization treatment at WWTP and the other types of sludge occurring in WW such as oils or gases, that recycling as soil amendments is not acceptable. According to the US EPA (Emerging Technologies for Biosolids Management, 2006), biosolids comprise solids, semisolids, or liquid residues created during WW treatment processes. Biosolids include solids removed during primary, secondary, or tertiary (advanced) treatment processes of WW or any substances derived from biosolids (Emerging Technologies for Biosolids Management, 2006; H. Wang et al., 2008).

Physical and chemical properties of biosolids mostly depend on technology and treatment process, also on the retention time in WW equipment. Composting, heat treatment, anaerobic digestion, and lime addition are stabilization technologies for biosolids (Sludge Stabilisation, 2011). The chemical composition of biosolids depends on the primary producers and varies with time and season. Biosolids, following the stage of mechanical dewatering, are masses presenting a lumpy structure and a bulk density within the range from 650 to 800 $kg \bullet m^{-3}$. The dry solid content of biosolids oscillates 2–12% by weight and is one of their most important parameters when waste-to-energy management is considered (Paz-Ferreiro et al., 2018; Fytili and Zabaniotou, 2008). Biosolids consist mainly of OM (usually more than 50% of the dry matter), mostly hydrocarbons, amino-acids or lipids, and with a small presence of lignin or cellulose (Organic contaminants in sewage sludge for agricultural use, 2001). The content of OM in biosolids from cities is higher but varies according to the treatment and conditioning carried out on the biosolids. It may be reduced as an effect of dilution, e.g., by lime addition. OM presented in biosolids can lead to improvements in the physical properties of soil, including the soil structure and attenuating the potential for surface runoff and erosion (Paz-Ferreiro et al., 2018).

Biosolids are particularly rich in N (2–8%), P (1.5–3.5%), and K (0.1–0.6%) (L. Wang et al., 2008; Sharma et al., 2017; Sullivan et al., 2015; Lu et al., 2012). Other nutrients found in biosolids are S (0.6–1.4%), Ca (1.0–4.0%) and Mg (0.4–0.8%) (Sullivan et al., 2015; Lu et al., 2012). Nutrient values in biosolids vary according to the wastewater source and wastewater treatment processes, such as dewatering and stabilization processes. Processes also affect the rate of nutrient release (or mineralization) (Lu et al., 2012).

Biosolids are a rich source of P, as without its coagulation at the WWTP, excessive P would be released into the environment. Phosphorus is one of the elements responsible for eutrophication and it is important to control its cycling in the environment (Carey and Migliaccio, 2009). The bioavailability of P is not dependent on its total concentration but is related to the presence of Al, Fe, Ca, and Mg cations in the input material (Lu et al., 2012). P occurs in both organic and inorganic compounds. Inorganic P is the predominant form, accounting for 70–90% of the total. Most of the P biosolids produced by the chemical treatment of wastewater during phosphorus removal are associated with inorganic phosphates bound to Fe or Al (TORRI et al., 2017). It is reported that the available phosphorus represents 20–60% of the total phosphorus in the biosolids (Sullivan et al., 2015). However, the mineral form of phosphorus captured in the sludge is hardly available to plants (O'Connor et al., 2004). Phosphorus is released into the soil solution when phosphorus-containing minerals dissolve and is taken up by plants in this form. The forms of phosphorus dissolved in the soil solution under normal pH conditions include the negatively charged primary orthophosphate anion ($H_2PO_4^-$) and a smaller amount of secondary orthophosphate anion (HPO_4^{2-}). When comparing fertilizer efficiency and phosphorus application to the soil, it is necessary to distinguish between elemental phosphorus and phosphate (P_2O_5), common commercial fertilizers are given on the basis of P_2O_5 (Mullins and Thomason, 2019).

Nitrogen is also abundant in biosolids. N occurs in both organic and inorganic (ammonium (NH_4^+), nitrites (NO_2^-), and nitrate (NO_3^-)) compounds (Rigby et al., 2016; Silva-Leal et al., 2021). Dewatered anaerobically digested biosolids contain about 20% of total N in the ammonium form, and 80% of total N in the organic form (Sullivan et al., 2015). SS treatment, dewatering processes, and stabilization processes also affected the proportion of mineral and organic N in the biosolids, as well as P (Rigby et al., 2016). Ammonium-N is available to the plant immediately after application, whereas organic N provides slow-release nitrogen because it must first be converted to ammonium-N by microbial processes (Sullivan et al., 2015).

3. Pyrolysis process

Different methods and parameters of biochar production, such as heating temperature, rate, and residence time, have found that the properties of biochar can be changed depending on the production technology. The production process is started by the burning of fuels, by electrical heating, or by microwaves. At present, cheap, simple, and traditional charcoal-making technologies are being switched over to modern biochar production technologies that result in greater biochar yield (Ok et al., 2015). The most commonly used method for biochar production nowadays is pyrolysis. However, relevant technological innovation is required to deal with environmental issues such as smoke production from traditional biochar production and associated health risks (Sohi et al., 2009).

3.1. Pre-treatment process

3.1.1. Mixing with additives

A catalyst is a substance that enters a chemical reaction, which by its presence accelerates or slows it down until an equilibrium state is reached. Its characteristic feature, however, is that it remains unchanged after the reaction. By adding the right catalyst to the material being processed, very efficient material selectivity and associated higher yields of the gaseous or liquid component can be achieved (Huang et al., 2016; Yu et al., 2014; Raček et al., 2022). Catalysts such as HZSM-5 (zeolite), H_3PO_4, NaCl, NaOH, etc. have shown increased yields of the solid output component, biochar, and reduced gas yields on other processed materials. The production of the liquid component remained essentially unchanged. However, these catalysts caused a decrease in CH_4 and CO_2 production. (Huang et al., 2016).

3.1.2. Pelletizing

Pelletizing is used to rebuild or transform bulk materials to be suitable for further processing (as feedstock for the pyrolysis process). The content of dry matter in the input material plays an important role in pyrolysis, while the optimal dry matter content is around 88% (Ševčík et al., 2018). This content is achieved by dehydration and subsequent heat drying, or a combination of both processing steps. The technology of pelletizing is used for the production of granules with a diameter of 3–20 mm. The principle of pelletizing is based on the use of a flat die, where the product is pressed through holes in the matrix by grinding rolls and transformed into endless strands, which are subsequently cut to the desired length by rotating knives. The mill rolls and the perforated die are the main components of the pelletizing process. The compaction takes place in the open active holes of the matrix. The material is fed to the pressing nut most often by a screw or belt conveyor. The final granulation takes place under the combined action of pressure, heat, and moisture. It is necessary that feedstock is converted into granulates that can be compacted for homogenization. The pellets are hot, soft, and sticky after discharge from the die matrix. The processing of municipal organic wastes and industrial biosolids into pellets ensures dust-free transport and increased product efficiency for further heat recovery (Racek et al., 2020).

3.2. Thermal treatment processes

Thermochemical treatment, also referred as a dry treatment process, uses temperature to break chemical bonds in biosolids. In general, there exist many thermochemical treatment processes, which can be classified into two main categories, such as thermochemical oxidative processes and thermochemical reductive processes. Thermochemical oxidative treatment processes include combustion, in which oxygen is present in stoichiometric or larger amounts. Thermochemical reductive processes include gasification, hydrothermal carbonization, and pyrolysis, in which oxygen is absent or the amount of oxygen is sub-stoichiometric (Šejvl, 2009). The classes of the thermochemical reductive treatment process of biosolids are focusing on the process temperature, residence time (some authors report retention time or process time), and products in weight percentage. The thermochemical oxidative process combustion incinerates wastes, biomass, and biosolids and converts them into ash, flue gas, and heat (Quan et al., 2022). The ash consists mostly of inorganic parts of the waste, which are formed by solid particles such as slag or particulates carried by the flue gas. Hazardous gaseous and particulate pollutants must be removed from the flue gasses before releasing into the atmosphere. Depending on the composition of the waste material, such as biosolids, combustion can be carried out by incineration and co-incineration, resulting in a reduction of the solid part of the original waste by more than 80%. Combustion of biosolids with respect to the water content can be incineration or co-incineration with other waste at the municipal solid waste incinerator (Gleau et al., 2017). Generally, combustion is an option for the valorization of hazardous waste such as biosolid materials into energy, while the high temperature of 850–1200 °C can destroy pathogens and toxins (Racek et al., 2020; Integrated Pollution Prevention and Control, 2006; Johnke, 2003).

3.2.1. Thermal pyrolysis

Pyrolysis conditions and feedstock materials have strong effects on the characteristics of biochar. The pyrolysis process greatly affects the qualities of biochar and its potential value to agriculture, either in terms of agronomic performance or carbon sequestration. The process and its parameters, principally temperature and furnace residence time, are particularly important, however, the process and process conditions also interact with feedstock type in determining the nature of the product (Ok et al., 2015). Pyrolysis is the thermochemical reductive process of converting biosolids into a carbon-rich solid product, biochar, with a yield above 10–80% car-

ried out under a wide temperature range of 100–1300 °C at a wide residence time 0,05 s–12 h under low or atmospheric pressure (Thangarajan et al., 2015; Callegari and Capodaglio, 2018).

The other products of pyrolysis are pyrolysis oil, thus bio-oil mixed with water, and pyrolysis gas, sometimes called syngas. Pyrolysis gas mainly contains CO, CO_2, CH_4 and H_2 (Ševčík et al., 2018). In pyrolysis, the organic compounds are converted into stable low-molecular substances and a solid carbonaceous residue. In the case of biosolids, the following polymeric substances are decomposed: cellulose, hemicellulose, and lignin. Many waste materials can be used as feedstock for pyrolysis: biosolids, green waste, sawdust, waste wood, nut shells, straw, cotton trash, rice hulls, switch grass, poultry litter, cattle manure, PW, paper sludge, and many more (Ghodake et al., 2021).

Depending on the residence time, pyrolysis can be divided into three classes: flash, fast, and slow pyrolysis (Callegari and Capodaglio, 2018). The yields of individual pyrolysis products were compared, the highest level of pyrolysis gas yield was observed for flash pyrolysis (40%), and torrefaction (20–40%), the highest level of pyrolysis oil yield was achieved for fast pyrolysis (50–70%), and the highest level of biochar yield was achieved for flash pyrolysis (60%), and torrefaction (60–80%). Depending on the process temperature, pyrolysis can be divided into three classes: low-temperature pyrolysis (< 500 °C), medium-temperature pyrolysis (500–800 °C), and high-temperature pyrolysis (> 800 °C) (Racek et al., 2020; Ochodek et al., 2007).

3.2.2. Microwave torrefaction

Selected literature suggests torrefaction, which can be described as mild pyrolysis, occurs when biosolids are heated in an inert atmosphere at a temperature of about 200–300 °C for a residence time of 30–240 min. Torrefaction produces primarily 60–80% solids, 20–40% gases, and a relatively small amount of liquid that can be burned together with gases (Racek et al., 2020). Torrefaction removes water as well as superfluous volatiles and partly decomposes the biopolymers (cellulose, hemicelluloses, and lignin) by giving off organic volatiles. It tends to yield higher proportions of solid (torrefied feedstock) in addition to liquid and non-condensable gaseous products. As specified by the EBC (European Biochar Certificate, 2022), the molar oxygen to carbon (O/C) ratio of biochar should be less than 0.4. But torrefied feedstock tends to have a higher O/C ratio than the EBC specification of biochar, therefore, the torrefied feedstock cannot be referred to as biochar. In that case, torrefaction is only a pre-treatment method that is used primarily for moisture removal and densification of feedstock, which will reduce the cost of transportation and increase the heating value of the feedstock. Torrefaction also increases the hydrophobicity, grindability, and reduces the biodegradability compared to the untreated feedstock. Torrefied feedstock can be stored long-term without degradation. A typical mass yield from torrefaction is 70–80% and energy yields are 80–90% (Daful & R Chandraratne, 2020).

4. Biochar

4.1. Biochar properties

When applying biochar, its properties must be evaluated at least according to the European Biochar Certificate (EBC) (European Biochar Certificate, 2022). The basic properties analyzed include the elemental composition of the biochar, pH, specific surface area, water holding capacity, electrical conductivity, bulk density, water, and dry matter content. Selected characteristics of biochar made from biomass and biosolids are noted in Table 2.

4.1.1. Organic carbon

One of the most important parameters of the biochar is the carbon (C) content. The C content depends mainly on the temperature of the pyrolysis process and the input feedstock (Ok et al., 2015). At temperatures above 600 °C, carbonization takes place, thus removing all remaining non-carbon atoms and increasing the C concentration in the biochar (Lehmann and Joseph, 2009). The C content ranges from 35 to 95% depending on the type of material (Ok et al., 2015; European Biochar Certificate, 2022). Pyrolyzed straw reaches values between 40 and 50% and shells between 70 and 90% (European Biochar Certificate, 2022). The C content of pig manure biochar is between 40 and 60% based on several studies (Tsai et al., 2012; Brassard et al., 2014; Sarfaraz et al., 2020; Gondek and Mierzwa-Hersztek, 2016). According to Domingues et al. (2017) the content of C in biochar produced from chicken manure de-

Table 2
Chemical composition of biochar obtained by pyrolysis from different feedstocks.

Feedstock	Max. process temperature (°C)	C (%)	H (%)	N (%)	O (%)	pH (–)	Ash (%)	SSA ($m^2 \cdot g^{-1}$)	Authors
Biosolids	600	21.80	1.10	3.20	7.90	–	–	92.30	Xu et al. (2018)
Food waste	600	50.41	1.26	1.86	9.44	–	36.94	359.52	Liu et al. (2020)
Printing leaflets	600	24.10	0.70	–	1.95	10.00	74.90	–	Xu et al. (2017)
Paper sludge	650	41.35	2.05	3.52	8.98	10.31	–	87.18	Pariyar et al. (2020)
Wheat straw	600	73.40	1.85	4.62	11.70	10.50	8.31	144.00	Břendová et al. (2017)
Maize straw	600	29.00	1.07	1.22	23.90	10.20	44.90	105.00 (550 °C)	Břendová et al. (2017)
Barley straw	400	71.50	4.70	1.30	22.30	8.02	–	–	Kang et al. (2018)
Rice straw	700	76.21	0.59	0.60	15.62	10.50	–	225.00	Liu et al. (2016)
Walnut shell	550	48.98	6.10	0.38	44.54	–	3.72	–	Diao et al. (2021)
Fishmeal	650	50.06	–	2.02	29.24	–	–	33.89	Shikhaliyev et al. (2018)
Chicken manure	550	38.27	–	–	20.79	–	–	12.25	Jung et al. (2017)
Pig manure	600	41.10	0.80	2.50	17.60	10.34	47.50	3.40	Tsai et al. (2012)

creases with increasing temperature, a similar trend can be observed in the case of Jung et al. (2017). Table 2 shows the percentage C content of various feedstock.

4.1.2. Phosphorus

Phosphorus (P) belongs to the macrobiogenic elements and is one of the most important elements in plant nutrition (Kirkby and Johnston, 2008). Phosphorus is taken up by plants in the form of trihydrogen phosphoric acid anions, mainly in the form of $H_2PO_4^-$ and HPO_4^{2-} (Gul and Whalen, 2016; Glaser and Lehr, 2019). P is extremely chemically reactive and always exists as phosphate mineral in nature. P in soil is the most immobile and unavailable of all nutrient elements. The physicochemical properties of soils influence the forms of available P and its solubility. Important soil properties that affect solubility are pH, and concentrations of Fe, Al, and Ca (Holford, 1997). The amount of P depends on the type of biochar feedstock, and the results show the following distribution when comparing the P content: biosolids > animal waste > biowaste > crop residues (Glaser and Lehr, 2019; Li et al., 2019).

4.1.3. Molar ratio H/C and O/C

The H/C molar ratio indicates the degree of carbonization, stability of biochar (European Biochar Certificate, 2022), and aromaticity of biochar (Xiao et al., 2016; Chen et al., 2021b). The H/C molar ratio value required by EBC (European Biochar Certificate, 2022) must be below 0.7. The H/C depends on the temperature of the pyrolysis process, with increasing temperature the atomic ratio decreased (Xiao et al., 2016; Chen et al., 2021b). According to Table 2 the H/C molar ratio is in the following order of feedstock groups: manure < straws < PW < DFW < FW.

The O/C molar ratio can be used to indicate the hydrophilic nature of biochar, which varies according to feedstock material and pyrolysis temperature (Hassan et al., 2020). Biochar with an O/C molar ratio of <0.2 is typically the most stable, possessing an estimated half-life of >1000 years; biochar with an O/C ratio of 0.2–0.6 has intermediate half-lives of 100–1000 years; biochar with an O/C ratio of >0,6 has a half-life in the order of <100 years (Leng et al., 2019; Spokas, 2014). The O/C molar ratio value required by EBC (European Biochar Certificate, 2022) should be below 0.4 which means approximately medium stability of the biochar. The O/C molar ratio is according to Table 2 in the following order of feedstock groups: manure < straws < PW < DFW < FW < shells.

4.1.4. Bulk density

Bulk density is the mass per unit volume of a collection of particles or pieces. It is not an intrinsic property of the material but depends on the size, shape, and compaction of the particles (Joseph et al., 2018). During the pyrolysis process, volatile substances are removed from the solid feedstock structure and form a porous biochar structure. The higher the porosity, the lighter the biochar per unit volume becomes. The bulk density of biochar decreases with the higher treatment temperature (Weber and Quicker, 2018; Mukherjee and Lal, 2013). Omondi et al. (2016) compared the bulk density of biochar in more than 100 studies with a focus on differentiation according to textural group, pyrolysis temperature, feedstock material, and biochar rate. Most studies showed a reduction in bulk density on the addition of biochar, in spite of a few studies indicating an increase in bulk density, which was attributed to the reorganization of particles as water drained to lower regions of the columns in lab experiments. The magnitude of the reduction in bulk density with respect to the used raw material of the biochar was found to be lower in the use of crop residue (8.5%) than in manure waste (5.3%).

4.1.5. pH

Most produced biochars are alkaline, however, biochar can be produced at almost any pH between 4 and 12 (Lehmann and Joseph, 2009). Biochar produced at higher temperatures (>400 °C) usually has a higher pH than low-temperature biochar (<400 °C) from the same feedstock (Singh et al., 2017). The application of biochar could increase the pH value of the soil. Many other factors depend on pH, such as CEC, ash content, nutrient availability, and more (Ding et al., 2016). Ji et al. (2022) report that the pH of wood biochar is lower than that of crop residue and organic wastes. Table 2 shows the percentage pH of various feedstock.

4.1.6. Cation exchange capacity

Cation exchange capacity (CEC) represents the amount of exchangeable cations (e.g., Ca^{2+}, Mg^{2+}, K^+, Na^+, NH^{4+}) (Weber and Quicker, 2018). CEC largely depends on the structure of biochar, i.e., porosity, specific surface area (Weber and Quicker, 2018), pH (Abdel-Fattah et al., 2015), and presence of carboxyl functional groups (Tan et al., 2017). The decrease in CEC with increasing temperature can be attributed to the loss of carboxyl functional groups during pyrolysis (Lehmann and Joseph, 2009; Liang et al., 2006). The difference in values also concerns the age of the biochar, fresh biochar shows minimum CEC values (Joseph et al., 2018; Lehmann and Joseph, 2009).

4.1.7. Specific surface area

The specific surface area (SSA) according to S_{BET} is an important characterization and comparison criterion for the physical structure of biochar. The S_{BET} is used to evaluate the gas adsorption data and generate a specific surface area result expressed in units of area per mass of sample ($m^2 \bullet g^{-1}$) (European Biochar Certificate, 2022). The SSA is most affected by the porosity and the feedstock material. As the temperature increases, the pore volume in the biochar increases, and thus the SSA also increases (Leng et al., 2021; Tomczyk et al., 2020). Pores can be divided into three groups according to the standard of the International Union of Pure and Applied Chemistry (IUPAC): micropores (<2 nm), mesopores (2–50 nm), and macropores (>50 nm) (Leng et al., 2021). The pyrolysis process degrades OM (cellulose, hemicelluloses, and lignin) and thus creates a porous biochar structure, this is especially true at higher pyrolysis temperatures, therefore, wood-biochar is offered as a better option with respect to higher SSA (Tomczyk et al., 2020; Cao and Harris, 2010).

4.2. Biochar certification

The certification of biochar and its commercial use is defined according to two main initiatives (European Biochar Certificate, 2022; International Biochar Initiative, 2015). Both certifications define biochar as a substance produced by thermal pyrolysis of biomass and biosolids under low oxygen conditions, allowing the use of biosolids and biomass from different sectors as the main feedstock.

The International Biochar Initiative (IBI) (International Biochar Initiative, 2015) standardized the definition and testing guidelines for biochar production further used as a soil amendment (IBI Biochar Standards) with the intent to provide stakeholders and commercial entities with the standards to identify certain qualities and characteristics of biochar when it is used as a soil amendment. The IBI Biochar Standards are designed to support the IBI Biochar Certification Program. Biochar can be made from a variety of feedstocks using a variety of different thermochemical conversion processes of feedstocks, and can possess many different attributes. The feedstock, whether unprocessed or processed, can be a combination of feedstock and diluents but may not contain more than 2% by dry weight of contaminants, and any diluents that constitute 10% or more by dry weight of the feedstock material must be reported as a feedstock component. Unprocessed feedstock types include biosolids, rice hulls, straw, etc., and processed feedstock include different sorts of manure, etc. IBI Biochar Standards identify three categories of tests for biochar materials: category A for basic utility properties (required for all biochar), category B for toxicant assessment, and category C for advanced analysis and soil enhancement properties.

The European Biochar Certificate (EBC) (European Biochar Certificate, 2022) is a voluntary industry standard in Europe and Switzerland, and it is mandatory when using biochar in agriculture. EBC defines six different biochar grades: “Feed”, “AgroOrganic”, “Agro”, “Urban”, “ConsumerMaterials” and “BasicMaterials”. For gaining the certificate, EBC criteria must be met regarding the biosolids feedstock, the production method, the properties of the biochar, and the way of application. For the purposes of this paper, we are only interested in groups “AgroOrganic” and “Agro” that meet all the requirements of the Regulation (EU) 2019/1009 laying down rules on the making available of fertilizer products on the EU market and amending Regulations (EC) No 1069/2009 and (EC) No 1107/2009 and repealing Regulation (EC) No 2003/2003 (Regulation 2019/1009, 2019). According to Positive list of permissible feedstocks for the production of biochar, it is possible to use SW and WS in both groups “AgroOrganic” and “Agro”. PW and FW can only be used as “Agro” if the plastic contamination does not exceed 1% (Positive list of permissible biomasses for the production of biochar, 2022). Pyrolysis of non-plant feedstock such as biosolids, livestock manure, manure containing biogas digestates can also produce valuable raw materials that could be used for the bioeconomy and climate protection. In this sense, it is planned to include these raw materials in the list of EBC raw materials following a key review of the product safety verification and conditions of use.

Selected parameters for biochar certification according to the guidelines of IBI (International Biochar Initiative, 2015) and EBC (European Biochar Certificate, 2022) are reported in Table 3, where the requirements are divided into biochar of category A and B (IBI) and “AgroOrganic” and “Agro” (EBC). For biochar of category A and “Feed” and “AgroOrganic” biochar the limits of selected parameters are stricter. The selected parameters are divided into main general parameters such as toxicant assessment, maximum allowed thresholds, and other parameters. The first selected parameters of toxicant assessment are polycyclic aromatic hydrocarbons (PAHs), polychlorinated dibenzodioxins or dibenzofurans (PCDD/Fs), polychlorinated biphenyls (PCBs), and heavy metals (HMs): As, Cd, Cr, Cu, Pb, Hg, Ni, and Zn. These selected parameters represent chemicals hazardous for agricultural use with respect to plants and organisms in the soil environment. The other parameters represent indicators that improve soil conditions. According to the guidelines, most other parameters require a declaration. Conversely, specific requirements are placed on the organic carbon C_{org}, the molar ratio $H:C_{org}$ and $O:C_{org}$.

5. Influence of biochar on the performance of substrate medium of green infrastructure

Biochar has some potential benefits for urban GI substrates. Generally, biochar additions can substantially increase soil fertility, improve soil physical properties, improve soil water holding capacity, support soil remediation, and remove GHG. The studies in the following chapters examine the effects of biochar on substrates compared to substrates without biochar.

5.1. Soil nutrition and fertility

Biochar made from organic waste materials is specific in that it contains many important elements for growing crops such as N, P, K, Ca and micronutrients, which should gradually release into the soil. This means that biochar may not contain significant amounts of nutrients immediately after application, but probably will in the long term (Hossain et al., 2011). The nutrient content of biochar depends on the raw feedstock materials and the pyrolytic conditions, e.g., temperature. Biochar derived from biosolids generally contains higher levels of N and P than those derived from wood- and straw-based feedstock (Hossain et al., 2020; El-Naggar et al., 2019). Furthermore, the N content of biochar decreases with an increase in the pyrolytic temperature due to the conversion of parts of amino acids into pyridine-N and pyrrolic-N (Hossain et al., 2020). The presence of biochar increases the CEC of the soil by up to 40%, which means participation in the retention of cations needed for plant nutrition. CEC values are mainly dependent on the feedstock and the temperature used in the pyrolysis process (Domingues et al., 2017). Increasing CEC in soil reduces nutrient leaching from the soil profile and increases nutrient availability to plant roots (El-Naggar et al., 2019). Nutrient leachability and subsequent infiltration of groundwater are also prevented, which could increase the use of fertilizers (Callegari and Capodaglio, 2018).

The results of a study according to Hossain et al. (2011) show that the concentrations of N and P in the form available for plant uptake are very low and depend on the process temperature. As the pyrolysis temperature increased from 300 °C to 700 °C, the total nitrogen content decreased by 55%. This is due to the volatilization and loss of NH_4–N and NO_3–N fractions as volatile substances containing nitrogen groups. For phosphorus, increasing temperature had a positive effect on its amount, which increased. This proves

Table 3
Selected parameters for biochar certification according to guidelines EBC and IBI (European Biochar Certificate, 2022; International Biochar Initiative, 2015).

	Selected parameter	Unit	Guidelines to biochar certification			
			International Biochar Initiative (IBI)		European Biochar Certificate (EBC)	
			Category A	Category B	Agro Organic	Agro/Urban/Consumer Materials
Toxicant assessment, maximum allowed thresholds	PAHs total	mg•kg^{-1} dry wt-mass	6	300	4	Agro: 6; Urban/Consumer Materials: Declaration
	PCDD/Fs		17	17	20	20
	PCBs[a]		0.2	1	0.2	0.2
	As		13	100	13	13
	Cd		1.4	20	0.7	1.5
	Cr		93	100	70	90
	Cu		143	6000	70	100
	Pb		121	300	45	120
	Hg		1	10[b]	0,4	1
	Ni		47	400	25	50
	Zn		416	7400	200	400
Other parameters	Moisture	%	Declaration		Declaration	
	C_{org}	%	≥60	≥30	Declaration	
	$H:C_{org}$	Molar ratio	0.7 maximum		0.7 maximum	
	Total ash	%	Declaration		Declaration	
	Total N	%	Declaration		Declaration	
	pH	–	Declaration		Declaration	
	Electrical conductivity	dS•m^{-1}	Declaration		Declaration	
	Liming[c]	% $CaCO_3$	Declaration		–	
	Particle size distribution	%	Declaration		–	
	$O:C_{org}$	Molar ratio	–		0.4 maximum	
	Total P and K	mg•kg^{-1}	Declaration		Declaration	
	Available P	mg•kg^{-1}	Declaration		Declaration	
	Total Ca and Mg	mg•kg^{-1}	Declaration		Declaration	
	Available Ca and Mg	mg•kg^{-1}	Declaration		Declaration	
	S_{BET}	m^2•g^{-1}	–		Declaration	

[a] PCDD/Fs in ng•kg^{-1}.
[b] methyl mercury 10 mg•kg^{-1} and inorganic mercury 40 mg•kg^{-1}.
[c] Liming if pH is above 7.

that phosphorus is associated with the inorganic sludge fraction. Also, the concentration of all micronutrients (K, Ca, Fe, Mg, and S) as total elements increased with increasing pyrolysis temperature (Hossain et al., 2011). The influence of pyrolysis temperature on the content of available nutrients N and P is also confirmed by Kookana et al. (2011). Biochar produced at 450–550 °C has a low content of available N, however, raw materials rich in N can retain up to 50% of the original N even at low pyrolysis temperatures compared to wood-based biochar (Kookana et al., 2011).

In their study, Chen et al. (2021a) achieved an increase in the phosphorus (P) and carbon (C) contents by adding biochar made from biosolids. Compared to natural soil, the increase in available nutrients was approximately 28–90% and the increase in total nutrients was approximately 28–115%. The increase in nutrients N and P in the soil is also attributed to the high porosity and adsorption of the biochar; due to the retention of rainwater in the pores, it/biochar minimizes the leaching of nutrients from the soil.

In the AdMaS Research Center, we test the use of biochar from biosolid in GR substrates. Innovative recycled materials used in growing medium are recycled bricks and biochar made from biosolid. Recycled materials used in the mixture were supplemented with standard materials of GR growing media such as expanded clay, spongilite, and peat. The addition of biochar was 9.5%. In our study, it turned out that biochar produced from mineral-rich sewage sludge tends to release nutrients when leached out by rain. However, nutrient concentrations in the GR system may increase again due to nutrients leaching from the biochar in the growing media in the GR by the rainfall. As confirmed by analyses of the GR outflow water, the growing medium containing biochar contains more nitrogen and phosphorus originating from the biochar, which was produced by the pyrolysis of nutrient-rich sewage sludge. The chemistry of the growing medium (nutrients and pH) and water retention capacity probably play a crucial role in vegetation development differences. Already 4 months after installation, a visible difference between the two modules can be seen in Fig. 3. Biochar adds nutrients to the substrate and retains water, thus supporting the growth of Sedum in the right module (Petreje et al., 2023).



Fig. 3. Comparison of green roof modules after 4 months from installation at the AdMaS Research Center, in Brno, the Czech Republic.; left module without added biochar; right module with biochar (Petreje et al., 2023). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

5.2. Water holding capacity

Biochar added to the GR substrate or used in agriculture increases water retention capacity (WHC). WHC is the ability of a material to hold and retain water, depending on the total pore volume, SSA, pore structure, feedstock used, and soil structure (Weber and Quicker, 2018; Bikbulatova et al., 2018). Increasing soil retention capacity with biochar is important for soils with a higher sand content, as water retention in these soils is low and water scarcity is critical for initial plant growth (Basso et al., 2013; Laghari et al., 2015). Water retention in the soil is also important for stabilizing the soil environment and preserving the biological functions of the soil. Biochar with a higher SSA has a significant effect on WHC, the SSA of biochar increases with increasing pyrolysis temperature (Weber and Quicker, 2018). The ability to retain water is related to the hydrophobicity of biochar. In biochar produced at a temperature above 400–500 °C, reactions occur that lead to a loss of hydrophobicity, and the biochar becomes hydrophilic and can bind water (Weber and Quicker, 2018; Kinney et al., 2012). Higher hydrophobicity can be caused by compressed air in the pores, and clogging the pores with tars and oils that have not evaporated at lower temperatures. The better results for biochar formed at higher temperatures are probably due to the presence of an extended nanopore structure providing physical water adsorption (Das and Sarmah, 2015). The hydrophobicity of fresh biochar is due to the fact that it usually has large molecules with a high density of H at the surface, however, some studies show that the aging of biochar may increase hydrophobicity (Mia et al., 2017; Edeh and Mašek, 2022), a similar effect can be achieved by oxidation of the biochar surface at contact with air and water (Basso et al., 2013).

Influence on the quality and quantity of substrates in order to achieve increased rainwater retention without redundant roof weight is a key feature in optimizing the benefits achieved by the installation of GR. The rate of WHC of GR depends on the amounts of biochar added. It is recommended for GRs to have WHC between 35% and 65% $w \bullet w^{-1}$ (Werdin et al., 2021). According to their study, Cao et al. (2014) added different amount of biochar made from green waste. The biochar dose was 0, 10, 20, 30, and 40% $w \bullet w^{-1}$. By applying the maximum amount of biochar, i.e., 40% $w \bullet w^{-1}$, the WHC increased by at least 74% of the GRs substrate, relative to the control GR with no biochar addition. Biochar also significantly reduced bulk density, with 40% biochar substrates able to have an additional 1.5 cm m^{-2} depth compared to the same weight as the control GR, which would further increase water retention. Beck et al. (2011) added only 7% $w \bullet w^{-1}$ of biochar made from rice hulls, pecan shells, WS, and coconut shells and found that water retention in the nearly saturated substrate increased by only 4.4% (Beck et al., 2011). During a four-year experiment, WS biochar significantly increased WHC by 25–28% relative to the control soil during the first two years, decreasing over the next two years to evaluate the control soil (Safaei Khorram et al., 2020).

Chen et al. (2018b) focused their study on biochar produced from biosolids by pyrolysis at 600 °C and proceeded with an experiment aimed at a mixture of different amounts of biochar (5, 10, 15, and 20%) with natural soil and local natural soil as a substrate to GR with an average thickness of 25 cm. The substrates were analyzed as five individual samples. The received results showed that the application of biochar made from biosolids may be an important amendment to GR substrate due to the increased WHC (up to 35% depending on the rate of biochar) and prolonged permanent wilting point, as well as water availability for plants. Authors indicate that the use of 15% biochar was an optimal rate for soil improvement and suggest that biochar has excellent potential as an appropriate amendment to GR in order to improve the ecological benefits via exerting effects on the moisture, temperature, and nutrients of roof substrates and further enhancing plant performance and altering the microbial community.

In the study by Chowdhury (2021) was examined the impact of a commercial wood biochar in enhancing urban roadway soil infiltration and associated changes in soil aggregation were evaluated over time through field-scale monitoring combined with laboratory analyses. This study investigated the effects of biochar on soil water field capacity. The study collected soil samples from two sites and amended them with 2% and 4% biochar. After five months, the WHC of the biochar-amended soils was significantly higher than the undisturbed soils. At the first site, the increase was 17.4% and 23.2% for 2% and 4% biochar-amended soils, respectively. At the second site, the mean WHC was 38.2% and 44.7% higher than undisturbed soils for 2% and 4% biochar amended soils, respectively. An increase in WHC was also achieved in samples taken after 15 months.

In the field study by Nakhli et al. (2021), the data showed that the biochar-amended soil was more effective at reducing peak and mean runoff rates than the control soil. Specifically, the biochar-amended soil reduced the peak and mean runoff rates by 85.2 ± 0.9% and 88.9 ± 0.5%, respectively, while the control soil reduced the peak and mean runoff rates by 61.4 ± 0.9% and 57.9 ± 0.7%, respectively. This was supported by the measurement of effective saturated hydraulic conductivity in the field. The laboratory column experiments also showed that the biochar-amended soil outperformed the control soil, reducing peak and mean runoff rates by 39 ± 3% and 54 ± 2%, respectively. This demonstrates that biochar amendment can significantly increase effective saturated hydraulic conductivity and reduce stormwater runoff.

5.3. Soil remediation

Soil remediation is a process in which pollutants come into contact with biochar in the soil and contaminants are adsorbed on its surface. By applying biochar to the soil, we can remove organic pollutants (antibiotics, pesticides, PAHs, PCBs, etc.), as well as inorganic substances known as heavy metals (Cu, Zn, Cd, Cr, Pb, Ni, Hg, etc.) (Wang and Wang, 2019; Wang et al., 2021; Nartey and Zhao, 2014; Yaashikaa et al., 2020). Biochar has many properties that support soil remediation, among them are, e.g., high surface area, high carbon content, aromaticity, stability, and CEC (Wang and Wang, 2019; Gelardi and Parikh, 2021; Oni et al., 2019).

The amount of biochar added to the soil for heavy metal remediation varies in the studies and accounts for 0.5–30% (He et al., 2019). Metal ions are strongly adsorbed to specific active sites containing phenolic and carboxyl functional groups on the surface of biochar (Nartey and Zhao, 2014). According to current research, the mechanisms of heavy metal removal by the addition of biochar can be attributed to electrostatic interactions, precipitation, and other reactions (Peng et al., 2011; Lu et al., 2012; Jiang et al., 2012) The incorporation of biochar into the soil increases the number of negative charges on the soil surface, due to the decreasing zeta potential and the increase in CEC (Peng et al., 2011). Therefore, the electrostatic attraction between positively charged heavy metals and soil is enhanced. The reason for metal precipitation is the increase in soil pH resulting from the addition of biochar (Lu et al., 2012). A number of studies have provided reliable data on the effectiveness of biochar in removing heavy metals from aqueous solutions and soil. Beesley and Marmiroli (2011) used biochar and significantly reduced concentrations of Cd and Zn. The results of this study showed a threefold reduction of Cd concentration and a 45fold reduction of Zn concentration. Three-year monitoring by Bian et al. (2014) of the effect of wheat straw biochar at application rates of 0, 10, 20, 30, and 40 t ha^{-1} had a positive effect on reducing the concentration of Cd and Pb. There was no significant difference between the application rates. The decrease in Cd concentration varied between years and ranged from 28 to 71%, and the decrease in Pb concentration was between 17 and 80%. Immobilization of the Pb and Cd also occurred due to cation exchange on the porous carbon structure.

Based on published data, the mechanism of organic pollutant removal can be described as surface adsorption and equilibrium distribution (partitioning). Adsorption refers in particular to surface interactions leading to the adhesion of pollutants to the surface of biochar, while sorption involves both surface adsorption and the equilibrium distribution of pollutant molecules in the micropores of biochar. Sorption takes place in relation to surface properties, such as SSA, micropore volume, and pore size (Kwon and Pignatello, 2005; Ennis et al., 2012).

Khan et al. (2015) found that the effectiveness of different biochar (application rate of 5%) was in the following order: peanut shell biochar (84%) > soybean straw biochar (70%) > rice straw biochar (55%) > biosolid biochar (36%), in terms of reducing PAH bioaccumulation in turnip (Khan et al., 2015). The influence of biochar from biosolids (application rate of 10%) according to Khan et al. (2013) can reduce the total bioaccumulation of PAH concentrations in lettuce grown in contaminated soils by 58–63%. Besides, adding 2% biochar to the soils reduces the uptake of PCBs (especially for di-, tri-, and tetrachlorobiphenyls) by plants, according to Wang et al. (2013), total PCB concentrations in the roots of B. chinensis and D. carota are decreased by 61.5–93.7% and 12.7–62.4%, respectively. Atrazine is one of the most common pesticides, it is used as an herbicide against dicotyledonous weeds in the cultivation of corn, sugar cane, soybeans, etc. The addition of biochar from wheat straw in an application rate of 1% to soil increased the efficacy of atrazine herbicide by 3.5 times and at the same time increased its persistence (Nag et al., 2011). However, dairy manure biochar was effective in immobilizing atrazine, and the effectiveness was enhanced with increasing incubation time and biochar rates. After 210 days, soils treated with the highest rate of 5% biochar showed more than a 77% reduction in atrazine concentrations (Cao et al., 2011).

5.4. Carbon sequestration

Soil carbon sequestration is a process in which CO_2 is removed from the atmosphere and stored in the soil carbon pool. Continued increases in atmospheric CO_2 and global temperatures may have a variety of different consequences for soil carbon inputs via controls on photosynthetic rates and carbon losses through respiration and decomposition (Yaashikaa et al., 2020; Galinato et al., 2011). Sequestration of carbon through the deposition of biochar in soil is one of the appropriate methods for mitigating the effects of climate change (Lehmann et al., 2006). At higher pyrolysis temperatures, a more stable biochar is formed, but nevertheless, the highest carbon sequestration has been found to be at 500 °C (Purakayastha et al., 2015). The stability of the biochar is affected by the H/C ratio; its low value (below 0.7) makes it difficult to decompose the biochar in the soil (Wijitkosum and Jiwnok, 2019; Li et al., 2019). The

removal of CO_2 in GR takes place at two levels: the first is the direct removal of CO_2 in the leaf area of plants by means of photosynthesis and carbon storage in the substrate and plants. The second level is indirect CO_2 reduction by saving energy for heating and cooling buildings (Liu and Li, 2012; Shafique et al., 2020). Not only GR but also grasslands in city parks have great potential for storing carbon in the soil (approximately 25–150 g•cm^{-2}•year^{-1}) (Shafique et al., 2020). According to Woolf et al. (2010), the global sustainable implementation of biochar can compensate for a maximum of 12% of current anthropogenic CO_2–C emissions. The optimal composition of the substrate and the thickness of the GR layer, along with a selection of plants have a very important role in the design of these structures. A properly designed GR with these parameters in mind will, in the future, meet the requirements for capturing and maximally storing carbon on roofs. Shafique et al. (2020) compared many studies focusing on carbon sequestration in GRs. The results show that GRs without the addition of biochar have a storage potential of 0.3–9.8 kg C•m^{-2}•year^{-1}. In the study by Chen et al. (2018a), 10% of the biochar was used in the GR growing medium at a substrate depth of 25 cm. According to the results, 9.33 kg C•m^{-2}•year^{-1} was deposited, and it was confirmed that biochar has a high potential for carbon storage due to its long-term stability (Chen et al., 2018a).

5.5. Biochar for the mitigation of greenhouse gas emissions

GHGs absorb solar energy and heat radiated from the earth's surface in the atmosphere. They capture them in our atmosphere and prevent this energy and heat from being released into space. There is a lot of GHG in the atmosphere naturally. The problem is that human activity adds a huge amount of them, which disproportionately increases the greenhouse effect, and contributes to global warming. The growing amount of GHG in the atmosphere means that average temperatures, including sea temperatures and ocean depths are rising due to a physical phenomenon called the greenhouse effect. The share of the most widespread emissions in the atmosphere is divided as follows: 79% CO_2, 11% CH_4, 7% N_2O, and 3% HFC (Overview of Greenhouse Gases, 2022). The European Union has committed itself to reducing its GHG emissions by 20% by 2020, 40% by 2030, and 80–95% by 2050 compared to 1990 ("EU greenhouse gas emissions", 2020).

Biochar in soil has the ability to effectively remove GHG (Lehmann and Joseph, 2009; Mukherjee and Lal, 2013; Oni et al., 2019), the main characteristics of biochar for GHG reduction include surface area, functional groups, pore size, pore space, elemental composition, pH, and sorption sites. The presence of functional groups such as carboxylic or hydroxyl groups on biochar surfaces can adsorb GHG, thus reducing their emissions when compared to unamended soil (Mandal et al., 2016b). Biochar made from wood has shown a better ability to mitigate GHG compared to biochar made from animal waste and biosolids. Biochar produced at higher temperatures (>600 °C) pyrolysis has been shown to be more efficient at removing N_2O (Li et al., 2019), this is mainly attributed to high-surface biochar, which provides more adsorption sites for nitrogen oxides, which reduces the reduction of these gases (Oni et al., 2019).

A very stable biochar will last in the soil for more than hundreds of years before it undergoes chemical and microbial decomposition, mainly due to its polycyclic aromatic structure (Lehmann and Joseph, 2009; Singh and Cowie, 2010). Biochar has been shown to have a very positive effect on GHG reduction, resulting in delayed CO_2, reduced N_2O, and synthesis gas production as a substitute for fossil fuels for heat or electricity production, reducing the amount of fuel for growing or irrigating, and reducing the production of nitrogen fertilizers with high GHG emissions (Singh et al., 2010; Wood and Cowie, 2004). Gelardi and Parikh (2021) compared observations from 88 studies that involved GHG mitigation, especially N_2O, and found that biochar in soil could remove N_2O by up to 32%, but unfortunately this effect was not long-lasting and after a year the removal was negligible. Woolf et al. (2010) estimated that the global implementation of biochar systems could reduce global GHG emissions by approximately 1.8 Gt CO_2 eq. per year, or 12% of current anthropogenic CO_2 eq., with a 50% reduction due to carbon sequestration, 30% from fossil fuel substitution, and 20% from eliminated CH_4 and N_2O emissions. Results in the study by Hu et al. (2022) show that GR amended with biochar can reduce air pollution. The data shows that the addition of biochar had a direct impact on the concentration of several different pollutants, including SO_2, CO_2, CO, and NO_2. GR with biochar can reduce the concentration of CO_2 by 7.3–28.8%, CO by 10.6–20.7%, SO_2 by 9.8–21.6%, and NO_2 by 2.1–11.5%.

6. Conclusion

This paper provides a review of the use of biochar made from biomass and biosolid as a substrate for GI. This review compares biochar from different types of feedstock, e.g., FW, paper and pulp production, straw waste, walnut shells, DFW, animal manure, and biosolids. The influence of the biochar on the physical and chemical properties of the substrate in the GI depends mainly on the properties of the biochar, which are determined by the raw feedstock, pyrolysis temperature, pH, amount of organic C, O:C ratio, H:C ratio, bulk density, porosity, SSA, CEC, etc. The reviewed studies show an improvement in the soil properties of substrates when using biochar in GI, especially fertility, WHC, soil remediation, and mitigation of GHG. Biochar can long-term sequester carbon and thus reduce CO_2 from the atmosphere and improve economic and environmental applications with a focus on sustainability. Due to its porous surface, biochar is suited for increasing the water-holding capacity of soils. A wide range of elements important for crop cultivation are present in biochar, such as N, P, K, and micronutrients. Using biochar for environmental management and hazardous pollution remediation is desirable, especially in cities in connection with car traffic (green parking lots). Further experiments need to be conducted to describe the effect of biochar feedstock on the properties of substrates. The transformation of organic solid waste by pyrolysis into biochar and its application in the GI of cities are becoming important topics of discussion in the EU, especially in the context of water management and the circular economy, i.e., circular waste management.

Author statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed.

We further confirm that the order of authors listed in the manuscript has been approved by all of us. We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Conceptualization: Michal Novotný, Mihajlo Marković, Jakub Raček, Tomáš Chorazy, Petr Hlavínek; Literature search and data analysis: Michal Novotný; Writing – original draft: Michal Novotný; Writing – review and editing: Mihajlo Marković, Jakub Raček, Milan Šipka, Tomáš Chorazy, Ivana Tošić, Petr Hlavínek; Supervision: Jakub Raček, Tomáš Chorazy, Petr Hlavínek.

Ethical approval

The authors declare that this declaration is not applicable.

Funding

This paper has been worked out with the financial support of the Ministry of Foreign Affairs of the Czech Republic - Department of Development Cooperation and Humanitarian Aid within the solution of project No. 22-PKVV-001 “Use of biochar as materially transformed waste for extensive green roofs”.

Availability of data and materials

The datasets generated during and/or analyzed during the current study are available in reference list.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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